

Indifference Curves and Consumer Theory

A Complete Guide

ECO 310 Midterm Review

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1 Introduction to Indifference Curves

Definition 1 (Indifference Curve). An *indifference curve* is the set of all consumption bundles (x_1, x_2) that provide the consumer with the same level of utility. Mathematically:

$$IC_u = \{(x_1, x_2) : u(x_1, x_2) = \bar{u}\}$$

where $\bar{u}$ is a constant utility level.

The consumer is “indifferent” between any two points on the same indifference curve—they derive equal satisfaction from either bundle.

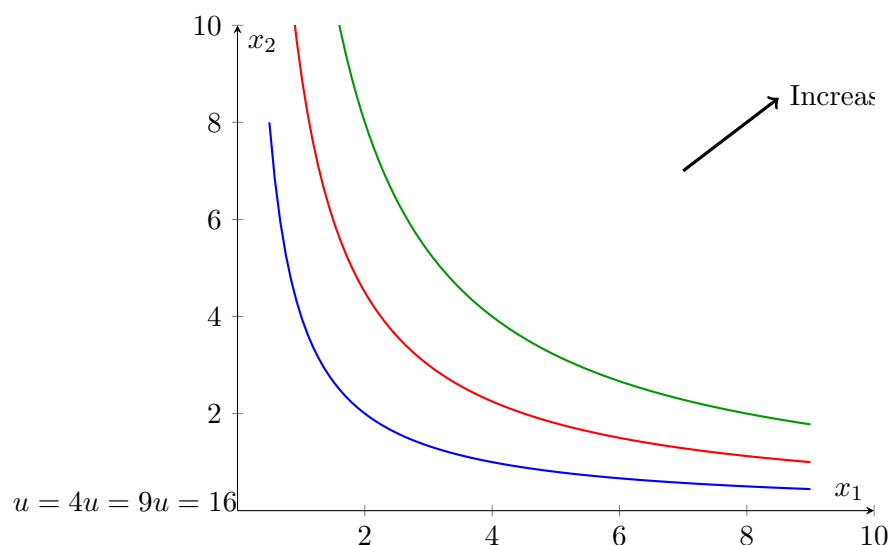
1.1 Key Properties of Indifference Curves

Property 1 (Downward Sloping). Indifference curves must slope downward (assuming both goods are desirable). To maintain constant utility, if you consume less of good 1, you must consume more of good 2 to compensate.

Property 2 (Higher Curves = Greater Utility). Indifference curves farther from the origin represent higher utility levels. This follows from the monotonicity assumption: more is better.

Property 3 (Never Cross). Two indifference curves representing different utility levels can never intersect. If they did, the point of intersection would simultaneously represent two different utility levels, which is impossible.

Property 4 (Convex to Origin). For most preferences, indifference curves are convex (bowed inward toward the origin). This reflects diminishing marginal rate of substitution.



2 Marginal Rate of Substitution (MRS)

Definition 2 (Marginal Rate of Substitution). The *MRS* at a point is the rate at which a consumer is willing to substitute good 2 for good 1 while maintaining the same utility level. It equals the absolute value of the slope of the indifference curve:

$$MRS_{12} = - \left. \frac{dx_2}{dx_1} \right|_{u=\bar{u}}$$

2.1 Computing MRS from Utility Functions

For a utility function $u(x_1, x_2)$, the MRS can be calculated as:

$$MRS_{12} = \frac{MU_1}{MU_2} = \frac{\partial u / \partial x_1}{\partial u / \partial x_2}$$

Derivation: Along an indifference curve, utility is constant, so $du = 0$. Using total differential:

$$\begin{aligned} du &= \frac{\partial u}{\partial x_1} dx_1 + \frac{\partial u}{\partial x_2} dx_2 = 0 \\ \frac{\partial u}{\partial x_1} dx_1 &= -\frac{\partial u}{\partial x_2} dx_2 \\ \frac{dx_2}{dx_1} &= -\frac{\partial u / \partial x_1}{\partial u / \partial x_2} = -\frac{MU_1}{MU_2} \end{aligned}$$

Therefore: $MRS_{12} = \frac{MU_1}{MU_2}$

Example 1 (Computing MRS). For $u(x_1, x_2) = x_1^{1/2} x_2^{1/2}$:

$$\begin{aligned} MU_1 &= \frac{1}{2} x_1^{-1/2} x_2^{1/2} = \frac{1}{2} \sqrt{\frac{x_2}{x_1}} \\ MU_2 &= \frac{1}{2} x_1^{1/2} x_2^{-1/2} = \frac{1}{2} \sqrt{\frac{x_1}{x_2}} \\ MRS_{12} &= \frac{MU_1}{MU_2} = \frac{x_2}{x_1} \end{aligned}$$

2.2 Diminishing MRS

Diminishing MRS means that as you consume more of good 1 (moving right along an indifference curve), you're willing to give up less and less of good 2 for additional units of good 1.

Economic Intuition: The more you have of something, the less valuable an additional unit becomes relative to what you're giving up.

Mathematical Implication: For convex preferences, MRS decreases as x_1 increases along an indifference curve.

3 Special Cases of Indifference Curves

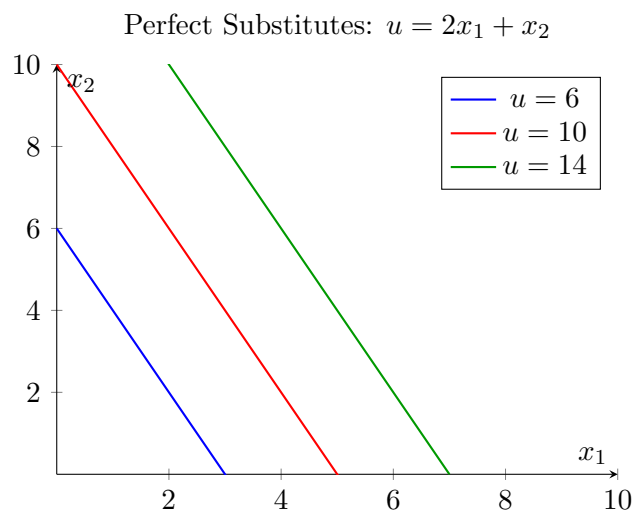
3.1 Perfect Substitutes

Definition: Goods are *perfect substitutes* if the consumer is willing to substitute one for the other at a constant rate.

Utility Function: $u(x_1, x_2) = ax_1 + bx_2$ where $a, b > 0$

MRS: $MRS_{12} = \frac{a}{b}$ (constant)

Indifference Curves: Straight lines with slope $-a/b$



Example: Nickels and dimes (2 nickels = 1 dime), different brands of identical products.

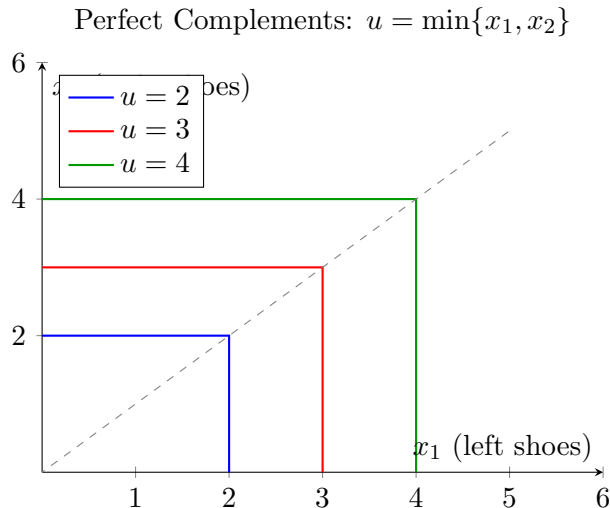
3.2 Perfect Complements

Definition: Goods are *perfect complements* if they must be consumed in fixed proportions to provide utility.

Utility Function: $u(x_1, x_2) = \min\{ax_1, bx_2\}$ where $a, b > 0$

MRS: Undefined at the corner; $MRS = 0$ on horizontal segment, $MRS = \infty$ on vertical segment

Indifference Curves: L-shaped (right angles)



Example: Left and right shoes, coffee and sugar (for some consumers), car engines and car bodies.

3.3 Cobb-Douglas Preferences

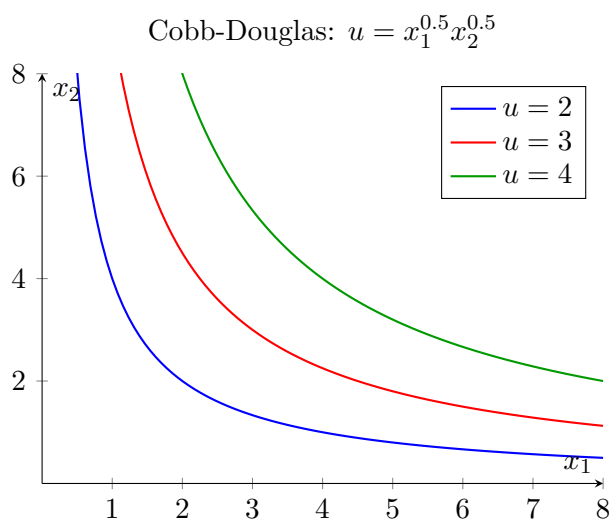
Utility Function: $u(x_1, x_2) = x_1^\alpha x_2^\beta$ where $\alpha, \beta > 0$

Often normalized so $\alpha + \beta = 1$: $u(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$

MRS:

$$MRS_{12} = \frac{\alpha x_1^{\alpha-1} x_2^\beta}{\beta x_1^\alpha x_2^{\beta-1}} = \frac{\alpha}{\beta} \cdot \frac{x_2}{x_1}$$

Indifference Curves: Smooth, convex hyperbolas (standard shape)



Key Property: Constant expenditure shares! Consumer spends fixed fraction of income on each good:

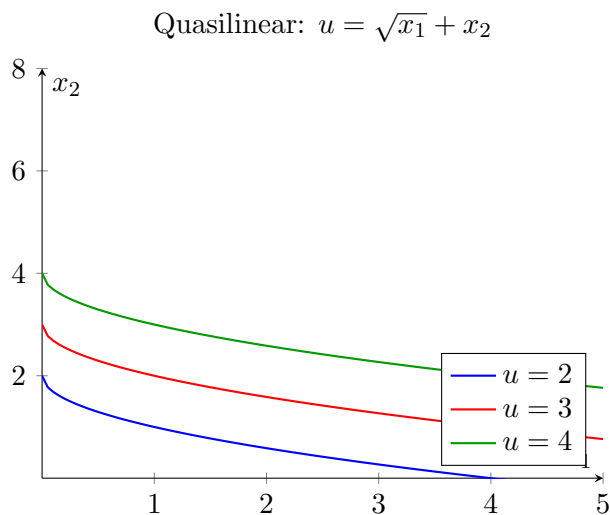
$$\frac{p_1x_1}{m} = \frac{\alpha}{\alpha + \beta}, \quad \frac{p_2x_2}{m} = \frac{\beta}{\alpha + \beta}$$

3.4 Quasilinear Preferences

Utility Function: $u(x_1, x_2) = v(x_1) + x_2$ where v is increasing and concave

MRS: $MRS_{12} = v'(x_1)$ (depends only on x_1 , not x_2 !)

Indifference Curves: Parallel vertical shifts of each other



Key Property: No income effects for good 1! Demand for x_1 depends only on prices, not income.

4 Consumer Optimization

4.1 Budget Constraint

The budget line shows all affordable bundles:

$$p_1x_1 + p_2x_2 = m$$

where m is income and p_1, p_2 are prices.

Slope of budget line: $-p_1/p_2$

4.2 Optimal Choice

Tangency Condition: At the optimal interior solution, the indifference curve is tangent to the budget line:

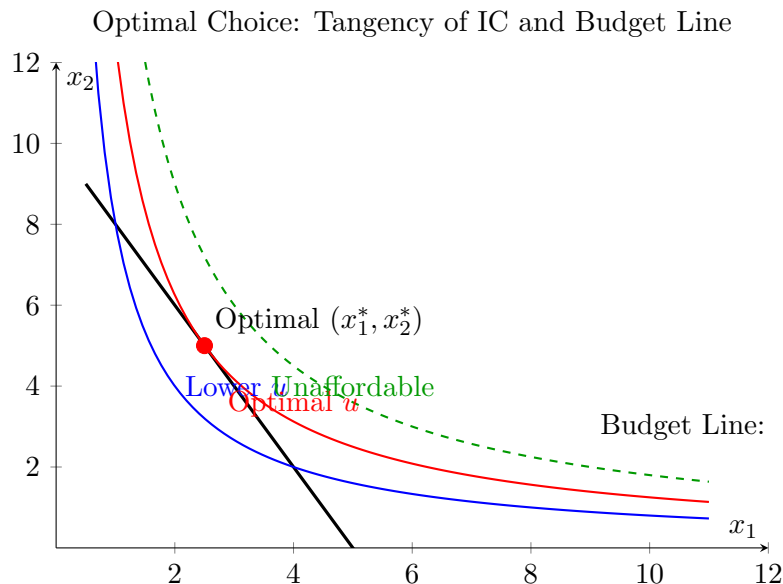
$$MRS_{12} = \frac{p_1}{p_2}$$

Interpretation: The rate at which the consumer is willing to trade goods (MRS) equals the rate at which the market allows trade (price ratio).

Full Optimality Conditions:

$$\frac{MU_1}{MU_2} = \frac{p_1}{p_2} \quad (\text{Tangency})$$

$$p_1x_1 + p_2x_2 = m \quad (\text{Budget constraint})$$

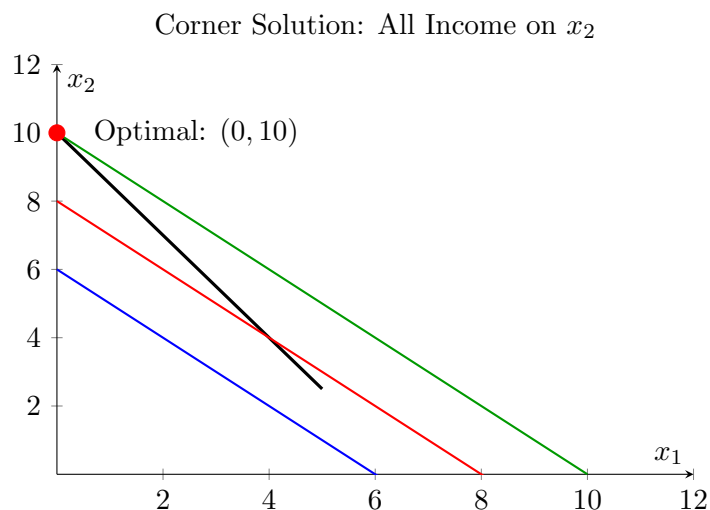


4.3 Corner Solutions

Sometimes the optimal choice is on the boundary (consumer chooses zero of one good).

Occurs when: $MRS \neq$ price ratio even at the boundary.

Example with Perfect Substitutes: If $MRS = a/b < p_1/p_2$, consumer buys only good 2.



5 Applications and Examples

5.1 Example 1: Deriving Demand (Cobb-Douglas)

Given $u(x_1, x_2) = x_1^{1/3} x_2^{2/3}$, $p_1 = 2$, $p_2 = 1$, $m = 30$.

Step 1: Tangency condition

$$\begin{aligned} MRS &= \frac{p_1}{p_2} \\ \frac{(1/3)x_1^{-2/3}x_2^{2/3}}{(2/3)x_1^{1/3}x_2^{-1/3}} &= \frac{2}{1} \\ \frac{1}{2} \cdot \frac{x_2}{x_1} &= 2 \\ x_2 &= 4x_1 \end{aligned}$$

Step 2: Budget constraint

$$\begin{aligned} 2x_1 + x_2 &= 30 \\ 2x_1 + 4x_1 &= 30 \\ x_1 &= 5 \end{aligned}$$

Step 3: Solve for x_2

$$x_2 = 4(5) = 20$$

Answer: Optimal bundle is $(x_1^*, x_2^*) = (5, 20)$.

Check: $\frac{p_1 x_1}{m} = \frac{2(5)}{30} = \frac{1}{3} \checkmark$ (matches Cobb-Douglas exponent!)

5.2 Example 2: Perfect Complements (Shoes)

Given $u(L, R) = \min\{L, R\}$ (left and right shoes), $p_L = 1$, $p_R = 4$, $m = 10$.

Key Insight: Always consume in fixed ratio: $L = R$ (need equal numbers)

Budget constraint with $L = R$:

$$\begin{aligned} p_L L + p_R R &= m \\ 1 \cdot L + 4 \cdot L &= 10 \\ 5L &= 10 \\ L &= 2 \end{aligned}$$

Answer: $(L^*, R^*) = (2, 2)$ — buy 2 pairs of shoes.

Note: If p_R increases to 5: - New budget: $L + 5L = 10 \Rightarrow L = 5/3$ - Demand for both decreases even though p_L didn't change! - This is the *income effect*

6 Summary: Convexity and MRS

Preference Type	MRS	Shape	Convex?
Perfect Substitutes	Constant	Straight line	Yes (weakly)
Perfect Complements	0 or ∞	L-shaped	Yes
Cobb-Douglas	$\frac{\alpha}{\beta} \frac{x_2}{x_1}$	Smooth hyperbola	Yes
Quasilinear	$v'(x_1)$	Parallel curves	Yes (if v concave)

Convex preferences are standard and well-behaved:

- MRS is diminishing
- Consumer prefers averages to extremes
- Unique optimal interior solution (usually)

When preferences are NOT convex:

- MRS may be increasing
- Consumer prefers extremes to averages
- Multiple optimal solutions possible
- Corner solutions more likely

7 Quick Reference: Computing MRS

General Formula:

$$MRS_{12} = \frac{MU_1}{MU_2} = \frac{\partial u / \partial x_1}{\partial u / \partial x_2}$$

Common Utility Functions:

1. **Cobb-Douglas:** $u = x_1^\alpha x_2^\beta$

$$MRS = \frac{\alpha}{\beta} \cdot \frac{x_2}{x_1}$$

2. **Perfect Substitutes:** $u = ax_1 + bx_2$

$$MRS = \frac{a}{b}$$

3. **Perfect Complements:** $u = \min\{ax_1, bx_2\}$

$$MRS = \begin{cases} 0 & \text{if } ax_1 > bx_2 \\ \infty & \text{if } ax_1 < bx_2 \\ \text{undefined} & \text{if } ax_1 = bx_2 \end{cases}$$

4. **Quasilinear:** $u = v(x_1) + x_2$

$$MRS = v'(x_1)$$

5. **CES:** $u = (x_1^\rho + x_2^\rho)^{1/\rho}$

$$MRS = \left(\frac{x_2}{x_1} \right)^{1-\rho}$$